TOPICS IN GROUP THEORY

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1. Groups, Subgroups, Homomorphisms, Quotients

Definition 1.1. — A *group*, denoted by pair $(\mathrm{G}, *)$, consists of a non-empty set G equipped with a *binary operation*, a map $*: \mathrm{G} \times \mathrm{G} \to \mathrm{G}, (x, y) \mapsto *(x, y)$ where we use the notation $x * y$ instead of $*(x, y)$, satisfying the following conditions :

(1) $x * (y * z) = (x * y) * z$ for all $x, y, z \in \mathrm{G}$.
(2) there exists some element $e \in \mathrm{G}$ such that $e * x = x * e = x$ for all $x \in \mathrm{G}$.
(3) for every $x \in \mathrm{G}$ there exists an element $y \in \mathrm{G}$ such that $x * y = y * x = e$.

The cardinality of a group is often called *order* of the group.

Exercise 1.2. — Show that the element e above is unique with respect to the property "$e * x = x * e = x$ for all $x \in \mathrm{G}$", that is to say : if there is yet another $e' \in \mathrm{G}$ satisfying "$e' * x = x * e' = x$ for all $x \in \mathrm{G}$" then $e = e'$. This e is called the *identity* of the group.

Exercise 1.3. — Show that given $x \in \mathrm{G}$, the element $y \in \mathrm{G}$ satisfying the property "$x * y = y * x = e$" is unique, that is to say : if $y' \in \mathrm{G}$ was another element satisfying $x * y' = y' * x = e$, then $y = y'$. This element $y \in \mathrm{G}$ for the given x is called the *inverse of* x and its is denoted by x^{-1}.

Definition 1.4. — Given an element $g \in (\mathrm{G}, *)$, we set

$$g^n := \begin{cases} g * \cdots * g \ (n\text{-times}) & \text{if } n > 0, \\ g^{-1} * \cdots * g^{-1} \ (n\text{-times }) & \text{if } n < 0, \\ e & \text{if } n = 0. \end{cases}$$

The least positive n for which $g^n = e$ is called *order of* g, denoted $o(g)$. If there does not exist any such positive integer we set $o(g) := \infty$.

Definition 1.5. — A group $(\mathrm{G}, *)$ is said to be *abelian* if $x * y = y * x$ for all $x, y \in \mathrm{G}$.

Example 1.6. — We give a few standard first examples of groups :

(1) The *trivial group* is the group with only one element, the binary operation being also trivial. It is often denoted by the *symbol* 1.
(2) Let X be any non-empty set, and consider the collection of bijections from X to X. This set is denoted by

$$\mathcal{S}_{\mathrm{X}} := \{f : \mathrm{X} \to \mathrm{X} \mid f \text{ is a bijection}\}.$$

Exercise. – Show that $\mathcal{S}_{\mathrm{X}}$ equipped with the binary operation of function composition is a group.

This group $\mathcal{S}_{\mathrm{X}}$ is called the *symmetric group on* X. When X is a finite set of n elements we use the notation $\mathcal{S}_n$ in place of $\mathcal{S}_{\mathrm{X}}$: this is called the *symmetric group on n letters.*
(3) The set of integers $\mathbf{Z}$ equipped with the binary operation of usual addition of integers is also a group, as can be checked easily.
(4) Similarly, the set of rational numbers $\mathbf{Q}$ or real numbers $\mathbf{R}$ or complex numbers $\mathbf{C}$, with the usual binary operation of addition are groups.
(5) The sets $\mathbf{Q}^{\times}$, $\mathbf{R}^{\times}$, $\mathbf{C}^{\times}$ equipped with the binary operation of addition *are not groups.* But equipped with the binary operation of usual multiplication do form groups.

(6) Let $n \geq 2$ and consider the matrices

$$\mathrm{R} = \begin{pmatrix} 0 & 0 & \cdots & 0 & 1 \\ 1 & 0 & \cdots & 0 & 0 \\ 0 & 1 & \cdots & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & 1 & 0 \end{pmatrix}, \quad \mathrm{S} = \begin{pmatrix} 0 & 0 & \cdots & 0 & 1 \\ 0 & 0 & \cdots & 1 & 0 \\ \vdots & \vdots & ⋰ & \vdots & \vdots \\ 0 & 1 & \cdots & 0 & 0 \\ 1 & 0 & \cdots & 0 & 0 \end{pmatrix}.$$

The set $\{\mathrm{R}^k, \mathrm{SR}^k \mid 0 \leq k < n\}$ forms a group, called the *dihedral group of order* $2n$, denoted by D_{2n}.

Exercise. – Show that $\mathrm{R}^n = \mathrm{I} = \mathrm{S}^2$ and $\mathrm{SRS} = \mathrm{R}^{-1}$. Using these show that the set $\{\mathrm{S}^i\mathrm{R}^j \mid i = 0{,}1 \text{ and } 0 \leq j < n\}$ is closed under matrix multiplication and has $2n$ many elements.

Definition 1.7. — Given a group $(\mathrm{G}, *)$, a non-empty subset H is called a subgroup if $e \in \mathrm{H}$ and $(\mathrm{H}, *)$ forms a group itself.

Exercise 1.8. — Let $g \in \mathrm{G}$. Show that $\{g^n \mid n \in \mathbf{Z}\}$ is a subgroup of G. This subgroup is denoted by $\langle g \rangle$ and it is called the (cyclic) *subgroup generated* by g.

Notation 1.9. — From now on, we will drop the symbol $*$ from the notation of an arbitrary group : that is to say, instead of writing phrases like "let $(\mathrm{G}, *)$ be a group..." we will simply write "let G be a group...". Also instead of writing $x * y$ for two elements $x, y \in \mathrm{G}$ we will simply write xy.

Definition 1.10. — Let G_1 and G_2 be two groups. A map $f : \mathrm{G}_1 \to \mathrm{G}_2$ is called a *homomorphism* if it satisfies : $f(xy) = f(x)f(y)$ for all $x, y \in \mathrm{G}_1$.

A homomorphism $f : \mathrm{G}_1 \to \mathrm{G}_2$ is called a *monomorphism* if it is injective, and it is called an *epimorphism* if it is surjective. A homomorphism of groups which is both injective and surjective is called an *isomorphism* of groups. Finally, an isomorphism from a group to itself is called an *automorphism*.

Exercise 1.11. — Let $f : \mathrm{G}_1 \to \mathrm{G}_2$ be a homomorphism of groups. Show that $f(e_{\mathrm{G}_1}) = e_{\mathrm{G}_2}$ where e_{G_i} for $i = 1{,}2$ denotes the identity of the group G_i.

Exercise 1.12. — Let $f : \mathrm{G}_1 \to \mathrm{G}_2$ be a homomorphism of groups. Show that $f(x^{-1}) = (f(x))^{-1}$ for all $x \in \mathrm{G}_1$.

Definition 1.13. — Let $f : \mathrm{G}_1 \to \mathrm{G}_2$ be a homomorphism of groups. The set

$$\operatorname{Ker} f := \{x \in \mathrm{G}_1 \mid f(x) = e_{\mathrm{G}_2}\}$$

is called the *kernel of* f.

Exercise 1.14. — Check that $\operatorname{Ker} f$ (equipped with the group operation of G_1) is a subgroup of G_1. Similarly, show that the image $\operatorname{Im} f$ is a subgroup of G_2.

Exercise 1.15. — Show that a group homomorphism $f : \mathrm{G}_1 \to \mathrm{G}_2$ is injective if and only if $\operatorname{Ker} f = \{e_{\mathrm{G}_1}\}$.

Definition 1.16. — A subgroup N of a group G is called *normal* if $gng^{-1} \in \mathrm{N}$ for every $g \in \mathrm{G}, n \in \mathrm{N}$.

Exercise 1.17. — Let $f : \mathrm{G}_1 \to \mathrm{G}_2$ be a group homomorphism. Show that $\operatorname{Ker} f$ is a normal subgroup of G_1.

Exercise 1.18. — Show that every subgroup of an abelian group is normal.

Theorem 1.19 (Cayley's theorem). — *Let* G *be a group. There exists a monomorphism of groups*

$$\varphi : \mathrm{G} \to \mathcal{S}_{\mathrm{G}}.$$

Thus, any group is isomorphic to a subgroup of some symmetric group.

Proof. — For any $g \in \mathrm{G}$ we will write φ_g instead of $\varphi(g)$. We define $\varphi : \mathrm{G} \to \mathcal{S}_{\mathrm{G}}$ as follows : for $g \in \mathrm{G}$ define $\varphi_g : \mathrm{G} \to \mathrm{G}$ as $x \mapsto gx$. We check that φ_g is a bijection. Indeed, given any $x \in \mathrm{G}$ we observe that $\varphi_g(g^{-1}x) = x$, i.e. φ_g is surjective. On the other hand, if $gx_1 = gx_2$ then multiplying g^{-1} of the left and using associativity we get $x_1 = x_2$; i.e. φ_g is injective as well.

Next, we check that $g \mapsto \varphi_g$ is a homomorphism. So given any $g_1, g_2 \in \mathrm{G}$ we must show that

$$\varphi_{g_1 g_2} = \varphi_{g_1} \circ \varphi_{g_2}.$$

Take any $x \in \mathrm{G}$ and observe that $\varphi_{g_1 g_2}(x) = (g_1 g_2)x = g_1(g_2 x) = \varphi_{g_1}(g_2 x) = \varphi_{g_1}(\varphi_{g_2}(x))$.

Finally, it remains to check that $g \mapsto \varphi_g$ is injective. Indeed, if $\varphi_{g_1} = \varphi_{g_2}$ then we have $\varphi_{g_1}(e) = \varphi_{g_2}(e)$ i.e. $g_1 = g_2$. The proof is complete. □

Remark 1.20. — In particular, any finite group G with n elements is isomorphic to a subgroup of $\mathcal{S}_n$.

Definition 1.21. — A group G *acts on a (non-empty) set* X if there is a map $\mathrm{G} \times \mathrm{X} \to \mathrm{X}$ (where the image of (g, x) is denoted by $g \cdot x$) satisfying

(1) $e \cdot x = x$ for all $x \in \mathrm{X}$.
(2) $g_1 \cdot (g_2 \cdot x) = (g_1 g_2) \cdot x$ for all $g_1, g_2 \in \mathrm{G}, x \in \mathrm{X}$.

Proposition 1.22. — *Group actions of* G *on* X *correspond to homomorphisms* $\mathrm{G} \to \mathcal{S}_{\mathrm{X}}$*.*

Proof. — Suppose $\mathrm{A} : \mathrm{G} \times \mathrm{X} \to \mathrm{X}$ is a group action. Consider the $\varphi_{\mathrm{A}} : \mathrm{G} \to \mathcal{S}_{\mathrm{X}}$ defined by $g \mapsto [x \mapsto g \cdot x]$. Then $\varphi_{\mathrm{A}}(g_1 g_2)(x) = (g_1 g_2) \cdot x = g_1 \cdot (g_2 \cdot x) = \varphi_{\mathrm{A}}(g_1)(\varphi_{\mathrm{A}}(g_2)(x))$ for all $x \in \mathrm{X}$. Therefore, we get $\varphi_{\mathrm{A}}(g_1 g_2) = \varphi_{\mathrm{A}}(g_1) \circ \varphi_{\mathrm{A}}(g_2)$ for all $g_1, g_2 \in \mathrm{G}$.

Conversely, assume $\varphi : \mathrm{G} \to \mathcal{S}_{\mathrm{X}}$ is a given homomorphism of groups. Define $\mathrm{A}_\varphi : \mathrm{G} \times \mathrm{X} \to \mathrm{X}$ by $(g, x) \mapsto g \cdot x := \varphi(g)(x)$. By Ex. 1.11 we know $\varphi(e) = \mathrm{id}_{\mathrm{X}}$ and so $e \cdot x = x$ for all $x \in \mathrm{G}$. Also, for $g_1, g_2 \in \mathrm{G}$ and $x \in \mathrm{X}$ we have

$$\begin{aligned}(g_1 g_2) \cdot x &= \varphi_{\mathrm{A}}(g_1 g_2)(x) \\ &= (\varphi_{\mathrm{A}}(g_1) \circ \varphi_{\mathrm{A}}(g_2))(x) \\ &= \varphi_{\mathrm{A}}(g_1)(g_2 \cdot x) \\ &= g_1 \cdot (g_2 \cdot x).\end{aligned}$$

□

Example 1.23. — Let G be a group. Consider the map $\mathrm{G} \times \mathrm{G} \to \mathrm{G}$ defined by $(g, x) \mapsto gx$. It is easy to see that this is a group action of G on itself. This is called the *left translation action.* Similarly, the map $\mathrm{G} \times \mathrm{G} \to \mathrm{G}$ given by $(g, x) \mapsto xg^{-1}$ is also an action, as can be verified. This is called the *right translation action.*

Exercise 1.24. — Verify that the map defined in the proof of Thm. 1.19 corresponds to the left translation action of G on itself.

Definition 1.25. — Let G be a group acting on a set X. Given $x \in \mathrm{X}$, the *orbit of* x is defined to be the set

$$\mathcal{O}_x := \{g \cdot x \mid g \in \mathrm{G}\}.$$

A set $\mathrm{T} \subset \mathrm{X}$ consisting of one element from each distinct orbit is called a *transversal.*

Proposition 1.26. — *Let* G *be a group acting on a set* X*. Then* X *is partitioned by orbits.*

Proof. — First we show that X is union of orbits. Indeed, given any $x \in \mathrm{X}$ we observe that $x \in \mathcal{O}_x$ since $e \cdot x = x$. Next we check that two orbits $\mathcal{O}_x$ and $\mathcal{O}_y$ are either equal or disjoint. Indeed, if $\mathcal{O}_x \cap \mathcal{O}_y \neq \emptyset$ then there is some $z \in \mathcal{O}_x \cap \mathcal{O}_y$. This means there are group elements $g_1, g_2 \in \mathrm{G}$ such that $g_1 \cdot x = z = g_2 \cdot y$. We show that $\mathcal{O}_x \subset \mathcal{O}_y$, and by symmetry $\mathcal{O}_y \subset \mathcal{O}_x$, whence $\mathcal{O}_x = \mathcal{O}_y$.

So take any $w \in \mathcal{O}_x$. Thus there exists $g \in \mathrm{G}$ such that $g \cdot x = w$. Now since $g_1 \cdot x = z = g_2 \cdot y$, we have $x = (g_1^{-1} g_2) \cdot y$ and thus we get

$$w = g \cdot x = g \cdot \left((g_1^{-1} g_2) \cdot y\right) = \left(g(g_1^{-1} g_2)\right) \cdot y \in \mathcal{O}_y.$$

Thus we have shown $\mathcal{O}_x \subset \mathcal{O}_y$. □

Notation 1.27. — Let G act on X. We denote the set of orbits by $\mathrm{G}\backslash\mathrm{X}$. The *class map* is denoted by

$$\pi : \mathrm{X} \to \mathrm{G}\backslash\mathrm{X}, \quad x \mapsto \mathcal{O}_x.$$

Thus X is partitioned into preimages of this class map.

Remark 1.28. — Note that the set of orbits $\mathrm{G}\backslash\mathrm{X}$ is a subset of $\mathcal{P}(\mathrm{X})$, the set of all subsets of X. Thus an orbit $\mathcal{O}_x$ can be thought of as an element of the set $\mathrm{G}\backslash\mathrm{X}$, and also as a subset of X. Therefore the partition of X obtained in the previous proposition is simply the partition of X as fibres of π.

Theorem 1.29 (Lagrange's theorem). — *Let* H *be a subgroup of a finite group* G*. Then* $|\mathrm{H}|$ *divides* $|\mathrm{G}|$*.*

Proof. — Consider the action $\mathrm{H}\times\mathrm{G}\to\mathrm{G}$ defined by $(h,g)\mapsto hg$. Then we have

$$\mathcal{O}_g = \{hg \mid h\in\mathrm{H}\}.$$

We will denote this orbit by Hg instead; this is often called the *right* H*-coset of* g. The cardinality of Hg is same as that of H : indeed, the map $h\mapsto hg:\mathrm{H}\to\mathrm{H}g$ is a bijection. Thus, since G is partitioned into orbits of the form Hg and each of these orbits have the same cardinality, that of H, we conclude that $|\mathrm{H}|$ divides $|\mathrm{G}|$. ☐

Exercise 1.30. — Show that $\mathrm{H}g=\mathrm{H}g'$ if and only if $gg'^{-1}\in\mathrm{H}$.

Exercise 1.31. — Let G be a group and H a subgroup. Verify that the map $\mathrm{H}\times\mathrm{G}\to\mathrm{G}$ defined by $(h,g)\mapsto gh^{-1}$ is an action of H on G. In this case the orbit of $g\in\mathrm{G}$ is denoted by gH, called the *left* H*-coset of* g. Also show that $g\mathrm{H}=g'\mathrm{H}$ if and only if $g^{-1}g'\in\mathrm{H}$.

Notation 1.32. — We denote the set of right H-cosets by $\mathrm{H}\backslash\mathrm{G}$ and the set of left H-cosets by G/H.

Recall that a subgroup N of a group G is called normal if $gng^{-1}\in\mathrm{N}$ for all $g\in\mathrm{G}, n\in\mathrm{N}$. If N is a normal subgroup of G we can define a binary operation on G/N by setting

$$(g_1\mathrm{N})\cdot(g_2\mathrm{N}) := g_1g_2\mathrm{N}.$$

We can verify that the above is well defined. Indeed, if $g_1\mathrm{N}=g_1'\mathrm{N}$ and $g_2\mathrm{N}=g_2'\mathrm{N}$ then $g_1^{-1}g_1'\in\mathrm{N}$. Thus $(g_1g_2)^{-1}(g_1'g_2') = g_2^{-1}(g_1^{-1}g_1')g_2\in\mathrm{N}$ i.e. $g_1g_2\mathrm{N}=g_1'g_2'\mathrm{N}$.

We can now check that G/N inherits a natural group structure from the group structure of G. In fact, eN is the identity and given any gN the inverse is g^{-1}N. This group G/N is called the *quotient group of* G *by* N.

Exercise 1.33. — Let G be a group. We call the set

$$\mathcal{Z}(\mathrm{G}) := \{g\in\mathrm{G}\mid gx=xg\ \forall x\in\mathrm{G}\}$$

the *center of* G. Show that $\mathcal{Z}(\mathrm{G})$ is a subgroup of G, and it is normal.

Exercise 1.34. — Show that the map $\mathrm{G}\times\mathrm{G}\to\mathrm{G},(g,x)\mapsto gxg^{-1}$ defines an action of G on itself. Also show that $\mathcal{Z}(\mathrm{G})=\bigcap_{x\in\mathrm{G}}\mathrm{Stab}_{\mathrm{G}}(x)$ with respect to this action.

Exercise 1.35. — Let $f:\mathrm{G}\to\mathrm{H}$ be a group homomorphism. Show that $\mathrm{Ker}\,f$ is a normal subgroup of G. Thus $\mathrm{G}/\mathrm{Ker}\,f$ has a natural group structure inherited from G. Show that the map

$$\mathrm{G}/\mathrm{Ker}\,f\to\mathrm{H},\quad g\mathrm{Ker}\,f\mapsto f(g)$$

is a well defined map and an injective homomorphism of groups. Thus conclude that we have an isomorphism of groups : $\mathrm{G}/\mathrm{Ker}\,f\simeq\mathrm{Im}\,f$. This is called the *fundamental theorem of isomorphism.*

Example 1.36. — Consider the group $\mathbf{Z}$. Then for any $n\in\mathbf{Z}$ the set $n\mathbf{Z}$ of multiples of n forms a subgroup under usual addition. In fact, we can show that any subgroup H of $\mathbf{Z}$ is of this form. Indeed, assume $\mathrm{H}\neq\{0\}$; then take n to be the least positive integer in H, and note that given any $m\in\mathrm{H}$ we can write $m=nq+r$ where $r\in\{0,1,\dots,n-1\}$. Thus it is forced that $r=0$: since $r=m-nq\in\mathrm{H}$ and n is the least positive integer in H.

Next, note that $n\mathbf{Z}$ is normal since $\mathbf{Z}$ is abelian. Thus $\mathbf{Z}/n\mathbf{Z}$ inherits a natural group structure prescribed by the group operation $+_n$ defined as

$$(x+n\mathbf{Z})+_n(y+n\mathbf{Z}) := (x+y)+n\mathbf{Z}.$$

In fact, we can define yet another binary operation $\cdot_n$ on $\mathbf{Z}/n\mathbf{Z}$ by setting

$$(x+n\mathbf{Z})\cdot_n(y+n\mathbf{Z}) = xy+n\mathbf{Z}.$$

We check the well-definedness : suppose $x-x'\in n\mathbf{Z}$ and $y-y'\in n\mathbf{Z}$; then we have

$$xy-x'y' = xy-x'y+x'y-x'y' = (x-x')y+x'(y-y')\in n\mathbf{Z}.$$

Thus $xy + n\mathbf{Z} = x'y' + n\mathbf{Z}$.

However, $\cdot$ is not a binary operation on the set of non-trivial cosets unless n is a prime number. Indeed, for instance if $n = pq$, a product of two distinct primes, then the product $(p + n\mathbf{Z}) \cdot_n (q + n\mathbf{Z}) = 0 + n\mathbf{Z}$, the trivial coset.

Now let $n = p$: a prime. Here $\cdot$ restricts to a binary operation on the set of non-trivial cosets $\{1 + p\mathbf{Z}, \ldots, p - 1 + p\mathbf{Z}\}$. The triple $(\mathbf{Z}/p\mathbf{Z}, +_p, \cdot_p)$ is called the *finite field of order p.* The word "field" is meant to indicate that $(\mathbf{Z}/p\mathbf{Z})^{\times} := \{1 + p\mathbf{Z}, \ldots, p - 1 + p\mathbf{Z}\}$ together with $\cdot_p$ is actually a group. It is clear that $1 + p\mathbf{Z}$ is the identity. It remains to check the existence of inverses. Indeed, suppose $i + p\mathbf{Z} \in (\mathbf{Z}/p\mathbf{Z})^{\times}$: then since i and p are coprime, there exists integers a, b such that $ai + bp = 1$. Thus $(a + p\mathbf{Z}) \cdot_p (i + p\mathbf{Z}) = ai + p\mathbf{Z} = 1 + p\mathbf{Z}$ i.e. $i + p\mathbf{Z}$ is invertible in $\left((\mathbf{Z}/p\mathbf{Z})^{\times}, \cdot_p\right)$.

Notation 1.37. — The finite field of order p is often denoted by $\mathbf{F}_p$.

Problems. We now look at some interesting problems concerning the examples encountered in the above section. In order to get the most out of the material, the reader should try to solve most of these in addition to the above routine exercises that are embedded in the main text.

Problem 1.38. — Let $s \in \mathbf{Z}$ and $n \in \mathbf{Z}_{>0}$. Set $\bar{s} := s + n\mathbf{Z} \in \mathbf{Z}/n\mathbf{Z}$. Show that the following properties are equivalent :

(1) s is coprime to n;
(2) $\bar{s}$ is a generator of the group $(\mathbf{Z}/n\mathbf{Z}, +_n)$ i.e. $\langle \bar{s} \rangle = \mathbf{Z}/n\mathbf{Z}$;
(3) $\bar{s} \in (\mathbf{Z}/n\mathbf{Z})^*$: the group of invertible elements for the ring multiplication $\cdot_n$ on $\mathbf{Z}/n\mathbf{Z}$.

Problem 1.39. — The groups $\mathbf{Z}/n\mathbf{Z}$ for $n \in \mathbf{Z}_{\geq 0}$ are called *cyclic groups* and they are often written as simply $\mathbf{Z}/n$.

(1) When $n = 0$ show that $\mathbf{Z}/0 \simeq \mathbf{Z}$.
(2) When $n = 1$ show that $\mathbf{Z}/1 \simeq 1$, the trivial group.
(3) When $n > 0$ show that $n\mathbf{Z} \simeq \mathbf{Z}$.
(4) Show that a subgroup of a cyclic group is also cyclic.
(5) We have already shown that subgroups of $\mathbf{Z}$ are of the form $n\mathbf{Z}$. Use the key idea of that proof to show that if $n > 1$ and $d \in \mathbf{Z}_{>0}$ is a divisor of n, then there exists precisely one subgroup of $(\mathbf{Z}/n\mathbf{Z}, +_n)$ of order d.

Problem 1.40. — Let G be any group and $\mathcal{Z}(\mathrm{G})$ be its center. Show that if $\mathrm{G}/\mathcal{Z}(\mathrm{G})$ is cyclic then G is abelian.

Problem 1.41. — Given $n \in \mathbf{Z}_{>0}$, we denote by $\varphi(n)$ the number of integers x such that $1 \leq x \leq n$ and x is coprime to n. This function $\varphi : \mathbf{Z}_{>0} \to \mathbf{Z}_{>0}$ is called the *Euler function.* Show the following :

(1) $\varphi(n) = |(\mathbf{Z}/n\mathbf{Z})^*|$.
(2) If p is a prime, then $\varphi(p) = p - 1$.
(3) If $\alpha \in \mathbf{Z}_{>0}$, then $\varphi(p^{\alpha}) = p^{\alpha-1}(p-1)$.
(4) There is a natural isomorphism of groups $\mathrm{Aut}(\mathbf{Z}/n\mathbf{Z}) \simeq (\mathbf{Z}/n\mathbf{Z})^*$, and in particular $\mathrm{Aut}(\mathbf{Z}/n\mathbf{Z})$ is an abelian group of cardinality $\varphi(n)$.

Problem 1.42. — Given two groups G_1 and G_2 one can define in a natural way a group operation on the Cartesian product $\mathrm{G}_1 \times \mathrm{G}_2$: for any $(g_1, g_2), (h_1, h_2) \in \mathrm{G}_1 \times \mathrm{G}_2$ we set $(g_1, g_2) \cdot (h_1, h_2) := (g_1 g_2, h_1 h_2)$. It is easy to check that this operation makes $\mathrm{G}_1 \times \mathrm{G}_2$ a group, called the *direct product of* G_1 *and* G_2. Also note that the projections $p_1 : \mathrm{G}_1 \times \mathrm{G}_2 \to \mathrm{G}_1$, $(g_1, g_2) \mapsto g_1$ and $p_2 : \mathrm{G}_1 \times \mathrm{G}_2 \to \mathrm{G}_2$, $(g_1, g_2) \mapsto g_2$ are group homomorphisms. Show that given any group Γ and homomorphisms $q_1 : \Gamma \to \mathrm{G}_1$, $q_2 : \Gamma \to \mathrm{G}_2$ there exists a unique homomorphism $\Phi : \Gamma \to \mathrm{G}_1 \times \mathrm{G}_2$ such that the triangles in the following diagram

$$\begin{tikzcd} & \Gamma \arrow[dl, "q_1"'] \arrow[d, dashed, "\Phi"] \arrow[dr, "q_2"] & \\ \mathrm{G}_1 & \mathrm{G}_1 \times \mathrm{G}_2 \arrow[l, "p_1"] \arrow[r, "p_2"'] & \mathrm{G}_2 \end{tikzcd}$$

commute – that is to say – $q_i = p_i \circ \Phi$ for $i = 1{,}2$.

Problem 1.43. — Generalize the above result to an arbitrary collection of groups $\{\mathrm{G}_i \mid i \in \mathrm{I}\}$ indexed by some indexing set I as follows. Put a natural groups structure on the Cartesian product $\prod_i \mathrm{G}_i$ and consider canonical projections $p_j : \prod_i \mathrm{G}_i \to \mathrm{G}_j$ for $j \in \mathrm{I}$. Show that given any group Γ with homomorphisms $q_j : \Gamma \to \mathrm{G}_j$ for $j \in \mathrm{I}$ there exists a unique homomorphism $\Phi : \Gamma \to \prod_i \mathrm{G}_i$ such that $q_j = p_j \circ \Phi$ for all $j \in \mathrm{I}$.

Problem 1.44. — Let p and q be distinct primes. Consider $\mathbf{Z}/p\mathbf{Z} \times \mathbf{Z}/q\mathbf{Z}$ with the direct product group structure as in the previous problem. Show that the map

$$n + pq\mathbf{Z} \mapsto (n + p\mathbf{Z}, n + q\mathbf{Z}) : \mathbf{Z}/pq\mathbf{Z} \to \mathbf{Z}/p\mathbf{Z} \times \mathbf{Z}/q\mathbf{Z}$$

is an isomorphism of groups.

Problem 1.45. — Let n be a positive integer, and write $n = p_1^{\alpha_1} \cdots p_r^{\alpha_r}$ where p_j are distinct prime factors of n and every α_j is positive. Show the following isomorphism of groups:

(1) $\mathbf{Z}/n\mathbf{Z} \simeq \prod_{j=1}^r \mathbf{Z}/p^{\alpha_j}\mathbf{Z}$;
(2) $(\mathbf{Z}/n\mathbf{Z})^* \simeq \prod_{j=1}^r (\mathbf{Z}/p^{\alpha_j}\mathbf{Z})^*$.

Now using Problem 1.41 show that

$$\varphi(n) = \prod_{j=1}^{r} \varphi(p^{\alpha_r}) = n \prod_{j=1}^{r} (1 - \frac{1}{p_j}).$$

Problem 1.46. — Consider the following matrices

$$a = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \quad \text{and} \quad b = \begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix}$$

in $\mathrm{GL}_2(\mathbf{C})$; here i denotes the square root of -1 in $\mathbf{C}$. Show that the set $\{a^i b^j \mid i, j \geq 0\}$ is closed under usual matrix multiplication. This group is called the *quaternion group* and denoted by $\mathbb{H}_8$. Show that $\mathbb{H}_8 = \{e, a, a^2, a^3, b, ab, a^2 b, a^3 b\}$ by showing that a has order 4, $b^2 = a^2$, and $ba = a^3 b$. Find all the subgroups of $\mathbb{H}_8$.

Problem 1.47. — Consider the following matrices

$$a = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \quad \text{and} \quad b = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

in $\mathrm{GL}_2(\mathbf{C})$. Show that $\{a^i b^j \mid i, j \in \mathbf{Z}_{\geq 0}\}$ is closed under matrix multiplication. Show that this group is isomorphic to D_8. Is it isomorphic to $\mathbb{H}_8$?

Problem 1.48. — Let G be a group whose every proper subgroup is cyclic. Is G necessarily cyclic, abelian? If we additionally assume G to be abelian, then is it cyclic?

Problem 1.49. — Let G be a group for which every subgroup is normal. Is G abelian?

Problem 1.50. — We have seen an example of the field $\mathbf{F}_p$. In general, a *field* is a triple $(\mathbf{F}, +, \cdot)$ consisting of a non-empty set $\mathbf{F}$ together with two binary operations $+$ and $\cdot$ such that

- $(\mathbf{F}, +)$ is an abelian group with the identity element denoted by 0,
- $(\mathbf{F} \setminus \{0\}, \cdot)$ is also an abelian group with identity denoted by 1,
- $a \cdot (b + c) = a \cdot b + a \cdot c$ and $(a + b) \cdot c = a \cdot c + b \cdot c$ for all $a, b, c \in \mathbf{F}$.

Let $\mathbf{F}$ be a field consisting of at least two elements. We define the *characteristic* of $\mathbf{F}$ to be

$$\operatorname{char} \mathbf{F} := \min \left\{ n \in \mathbf{Z}_{>0} \mid n \cdot 1 := \underbrace{1 + \cdots + 1}_{n\text{-times}} = 0 \right\}$$

if there exists some positive integer n such that $n \cdot 1 = 0$, or we set $\operatorname{char} \mathbf{F} := 0$. Show that $\operatorname{char} \mathbf{F}$ is either 0 or a prime number. Show that if $\mathbf{F}_q$ is a finite field of cardinality q, then q is power of some prime p which is precisely the characteristic of $\mathbf{F}_q$.

2. Sylow's Theorems

Definition 2.1. — Let G be a group acting on a set X. Given $x \in \mathrm{X}$ its *stabilizer* is defined as

$$\mathrm{Stab}_{\mathrm{G}}(x) := \{g \in \mathrm{G} \mid g \cdot x = x\}.$$

Exercise 2.2. — Check that $\mathrm{Stab}_{\mathrm{G}}(x)$ is a subgroup of G.

Theorem 2.3 (Orbit-Stabilizer theorem)**.** — *Let* G *be a group acting on* X*; fix an* $x \in \mathrm{X}$*. Then there is a bijection between* $\mathcal{O}_x$ *and* $\mathrm{G}/\mathrm{Stab}_{\mathrm{G}}(x)$*.*

Proof. — Define a map $g\mathrm{Stab}_{\mathrm{G}}(x) \mapsto g \cdot x : \mathrm{G}/\mathrm{Stab}_{\mathrm{G}}(x) \to \mathcal{O}_x$. If $g'\mathrm{Stab}_{\mathrm{G}}(x) = g'\mathrm{Stab}_{\mathrm{G}}(x)$, then by Ex. 1.31, we have $g^{-1}g' \in \mathrm{Stab}_{\mathrm{G}}(x)$, whence $g \cdot x = g' \cdot x$. Thus the map is well defined. It is clearly surjective. If $g\mathrm{Stab}_{\mathrm{G}}(x)$ and $g'\mathrm{Stab}_{\mathrm{G}}(x)$ are two cosets such that $g \cdot x = g' \cdot x$ then $g^{-1}g' \in \mathrm{Stab}_{\mathrm{G}}(x)$. Thus by Ex. 1.31 we have $g'\mathrm{Stab}_{\mathrm{G}}(x) = g'\mathrm{Stab}_{\mathrm{G}}(x)$. This proves the map defined above is injective. ☐

Proposition 2.4. — *Let* G *be a finite group, and let* p *be the smallest prime dividing the order of* G*. If* H *is a subgroup of* G *of index* p*, then* H *is normal in* G*.*

Proof. — Consider the action $\mathrm{H} \times \mathrm{G}/\mathrm{H} \to \mathrm{G}/\mathrm{H}$ given by $(h, g\mathrm{H}) \mapsto hg\mathrm{H}$. The orbit of the element 1H is $\{\mathrm{H}\}$, hence we have a decomposition $\mathrm{G}/\mathrm{H} = \{\mathrm{H}\} \sqcup \bigsqcup_{j=1}^{r} \mathcal{O}_{g_j\mathrm{H}}$. Thus $r \leq p-1$. By the Orbit-Stabilizer theorem, we know $|\mathcal{O}_{g_j\mathrm{H}}|$ divides $|\mathrm{H}|$ for each $j = 1, \ldots, r$. But by Lagrange's theorem $|\mathrm{H}|$ divides $|\mathrm{G}|$ and thus $|\mathcal{O}_{g_j\mathrm{H}}|$ divides $|\mathrm{G}|$ for every j. So if q is any prime divisor of $|\mathcal{O}_{g_j\mathrm{H}}|$ then $|\mathcal{O}_{g_j\mathrm{H}}| \geq q \geq p$. However, the union $\bigsqcup_{j=1}^{r} \mathcal{O}_{g_j\mathrm{H}}$ contains precisely $p-1$ many elements, and so every $\mathcal{O}_{g_j\mathrm{H}}$ must be a singleton. Hence, for any $g \in \mathrm{G}$ and $h \in \mathrm{H}$ we have $hg\mathrm{H} = g\mathrm{H}$ or equivalently $g^{-1}hg \in \mathrm{H}$, and so we conclude that H is normal. ☐

Exercise 2.5. — Let G be a finite group and p the smallest prime divisor of the order of G. If H is a normal subgroup of G of order p, show that $\mathrm{H} \subset \mathcal{Z}(\mathrm{G})$.
Hint : Consider the conjugation action of G on H and mimic the argument of the previous proof to show that every orbit is a singleton.

Example 2.6. — We now consider the vector space $\mathbf{F}_p^n$ over $\mathbf{F}_p$. We fix the standard basis

$$e_1, \ldots, e_n$$

where $e_i = (0, \ldots, 1, \ldots, 0)^{\intercal}$ has 1 in the i-th position. With respect to this standard basis we identify

$$\mathrm{GL}_n(\mathbf{F}_p) \simeq \mathrm{Aut}_{\mathbf{F}_p}\left(\mathbf{F}_p^n\right).$$

This group $\mathrm{GL}_n(\mathbf{F}_p)$ consists of invertible n-by-n matrices with entries from the field $\mathbf{F}_p$, or equivalently, matrices having non-zero determinant.

We can identify the group $\mathcal{S}_n$ as a subgroup of $\mathrm{GL}_n(\mathbf{F}_p)$. In fact, we can define a map

$$\varphi : \mathcal{S}_n \to \mathrm{GL}_n(\mathbf{F}_p), \quad \sigma \mapsto \mathrm{P}_\sigma := \begin{pmatrix} \vdots & \vdots & & \vdots \\ e_{\sigma(1)} & e_{\sigma(2)} & \cdots & e_{\sigma(n)} \\ \vdots & \vdots & & \vdots \end{pmatrix},$$

and we check that the above is an injective homomorphism. Indeed, first we have to show that

$$\mathrm{P}_\sigma \mathrm{P}_\tau = \mathrm{P}_{\sigma\tau}, \quad \forall \sigma, \tau \in \mathcal{S}_n.$$

To check this it suffices to apply both sides to standard basis vectors e_j. Observe that

$$\mathrm{P}_\sigma(\mathrm{P}_\tau(e_j)) = \mathrm{P}_\sigma(e_{\tau(j)}) = e_{\sigma(\tau(j))} = \mathrm{P}_{\sigma\tau}(e_j).$$

Next we check that φ is injective. If $\mathrm{P}_\sigma = \mathrm{P}_\tau$, then for each j we get

$$\mathrm{P}_\sigma(e_j) = e_{\sigma(j)} = e_{\tau(j)} = \mathrm{P}_\tau(e_j).$$

Therefore $\sigma(j) = \tau(j)$ for every j, hence $\sigma = \tau$.

Definition 2.7. — A finite group G of cardinality p^α where p is a prime and $\alpha \in \mathbf{Z}_{>0}$ is called a *p-group.*

Definition 2.8. — Let G be a finite group such that $|\mathrm{G}| = p^\alpha m$ where p is a prime, $\alpha \in \mathbf{Z}_{>0}$, and $p \nmid m$. A subgroup of G of order p^α is called a *Sylow p-subgroup.*

Example 2.9. — The group $\mathrm{GL}_n(\mathbf{F}_p)$ has cardinality

$$(p^n-1)(p^n-p)\cdots(p^n-p^{n-1}) = p^{\frac{n(n-1)}{2}}m$$

where $p \nmid m$. Now the upper unipotent subgroup $\mathrm{U}_n(\mathbf{F}_p)$ consisting of 1's in the diagonal and zeros below the diagonal has cardinality $p^{\frac{n(n-1)}{2}}$, and so it is a Sylow p-subgroup of $\mathrm{GL}_n(\mathbf{F}_p)$.

Lemma 2.10. — *Let* G *be a finite group and* S *be a Sylow* p*-subgroup of* G*. If* H *is any subgroup of* G *then there exists an element* $a \in \mathrm{G}$ *such that* $\mathrm{H} \cap a\mathrm{S}a^{-1}$ *is a Sylow* p*-subgroup of* H.

Proof. — Consider the action $\mathrm{H} \times (\mathrm{G}/\mathrm{S}) \to \mathrm{G}/\mathrm{S}$ defined by $(h, a\mathrm{S}) \mapsto ha\mathrm{S}$. Then for any $a\mathrm{S} \in \mathrm{G}/\mathrm{S}$ we observe that

$$\begin{aligned}\mathrm{Stab}_{\mathrm{H}}(a\mathrm{S}) =& \{h \in \mathrm{H} \mid ha\mathrm{S} = a\mathrm{S}\}\\ =& \{h \in \mathrm{H} \mid a^{-1}ha \in \mathrm{S} \text{ i.e. } h \in a\mathrm{S}a^{-1}\}\\ =& \mathrm{H} \cap a\mathrm{S}a^{-1}.\end{aligned}$$

Thus by Thm. 2.3 we know that $\left|\mathrm{H}/\left(\mathrm{H} \cap a\mathrm{S}a^{-1}\right)\right| = |\mathcal{O}_{a\mathrm{S}}|$. Now since G/S is a disjoint union of orbits $\mathcal{O}_{a\mathrm{S}}$ and $|\mathrm{G}/\mathrm{S}|$ is coprime to p, we conclude that there exists some orbit $\mathcal{O}_{a\mathrm{S}}$ which has cardinality coprime to p. For this choice of a the subgroup $\mathrm{H} \cap a\mathrm{S}a^{-1}$ is a Sylow p-subgroup of H. ☐

Theorem 2.11 (Sylow's first theorem)**.** — *Let* G *be a finite group of cardinality* $p^\alpha m$ *where* p *is prime,* $\alpha \in \mathbf{Z}_{>0}$*, and* m *is coprime to* p*. Then there exists a Sylow* p*-subgroup in* G.

Proof. — Set $n = p^\alpha m$; then embed $\mathrm{G} \hookrightarrow \mathcal{S}_n$ (Cayley's theorem) and $\mathcal{S}_n \hookrightarrow \mathrm{GL}_n(\mathbf{F}_p)$ (Eg. 2.6). Thus G can be identified as a subgroup of $\mathrm{GL}_n(\mathbf{F}_p)$. Since $\mathrm{GL}_n(\mathbf{F}_p)$ has a Sylow p-subgroup (see Eg. 2.9) we conclude by the previous lemma that G also has a Sylow p-subgroup. ☐

Lemma 2.12. — *Let* G *be a* p*-group acting on a finite set* X *and let*

$$\mathrm{X}^{\mathrm{G}} := \{x \in \mathrm{X} \mid g \cdot x = x \ \forall g \in \mathrm{G}\}.$$

Then we have

$$|\mathrm{X}| \equiv |\mathrm{X}^{\mathrm{G}}| \mod p.$$

Proof. — Note that $x \in \mathrm{X}^{\mathrm{G}}$ is equivalent to saying $\mathcal{O}_x = \{x\}$. Since X is partitioned into orbits, for a chosen transversal T, we have

$$|\mathrm{X}| = |\mathrm{X}^{\mathrm{G}}| + \sum_{x \in \mathrm{T}\setminus\mathrm{X}^{\mathrm{G}}} |\mathcal{O}_x|.$$

Now by Thm. 2.3, we know that $|\mathcal{O}_x|$ divides $|\mathrm{G}| = p^n$, say. Thus for each $x \in \mathrm{T} \setminus \mathrm{X}^{\mathrm{G}}$ we know that $|\mathcal{O}_x|$ is a positive power of p. Thus $\sum_{x \in \mathrm{T}\setminus\mathrm{X}^{\mathrm{G}}} |\mathcal{O}_x|$ is divisible by p. ☐

Exercise 2.13. — Consider the map

$$(g, x) \mapsto g \cdot x := gxg^{-1} : \mathrm{G} \times \mathrm{G} \to \mathrm{G}$$

which defines an action of G on itself, and show that with respect to this action G^{G} is precisely $\mathcal{Z}(\mathrm{G})$.

We now see an immediate application of the above lemma.

Lemma 2.14. — *The center of a* p*-group is non-trivial.*

Proof. — By the previous lemma and the exercise : $|\mathrm{G}| \equiv |\mathcal{Z}(\mathrm{G})| \mod p$. Since $|\mathrm{G}|$ is divisible by p, so is $|\mathcal{Z}(\mathrm{G})|$. Thus $|\mathcal{Z}(\mathrm{G})| > 1$. ☐

Exercise 2.15. — Let G be a group of order p^2 where p is a prime. Show that G is abelian.
Hint : Use Problem 1.40.

Theorem 2.16 (Sylow's first theorem – Stronger version)**.** — *If* $|\mathrm{G}| = p^\alpha m$ *with* $\alpha > 0$ *and* $p \nmid m$*, then* G *contains a subgroup of cardinality* p^i *for every* $i \leq \alpha$.

Proof. — It suffices to prove the statement for p-groups since we know by Thm. 2.11 that G has a subgroup P of cardinality p^{α}.

So let P be any p-group with $|\mathrm{P}| = p^{\alpha}$; if $\alpha = 1$, then there is nothing to prove. So we assume $\alpha > 1$ and that the statement of the theorem is true for p-groups of cardinality $p^{\alpha-1}$. Since $\mathcal{Z}(\mathrm{P})$ is non-trivial, we take some non-identity element $z \in \mathcal{Z}(\mathrm{P})$. By Lagrange's theorem we know that $|z| = p^r$ for some $r \geq 1$. Now take $\mathrm{N} := \langle z^{p^{r-1}} \rangle$; this is subgroup of $\mathcal{Z}(\mathrm{P})$ of cardinality p. Also being a subgroup of the center N is normal in P. So we consider the quotient group P/N which has cardinality $p^{\alpha-1}$. By induction hypothesis, the quotient group has a subgroup $\bar{\mathrm{H}}$ of order j for each $0 \leq j \leq \alpha - 1$. Let $\pi : \mathrm{P} \to \mathrm{P/N}$ be the canonical quotient map, and let H be the preimage of $\bar{\mathrm{H}}$ under π. Now note that N is obviously a subgroup of H and $\mathrm{H/N} \simeq \bar{\mathrm{H}}$. Therefore $|\mathrm{H}| = |\mathrm{N}| \cdot |\mathrm{H/N}| = p^{j+1}$. Consequently, P has subgroups H which are of cardinality p^j for all $j \leq \alpha$. □

Exercise 2.17. — Let H and K be subgroups of a group G. The *subgroup generated by* H *and* K is defined as the intersection of all subgroups of G containing $\mathrm{H} \cup \mathrm{K}$. We denote this subgroup by $\langle \mathrm{H}, \mathrm{K} \rangle$. Show that

$$\langle \mathrm{H}, \mathrm{K} \rangle = \{x_1 \cdots x_n \mid n \geq 1,\, x_i \in \mathrm{H} \cup \mathrm{K}\}.$$

Thus $\langle \mathrm{H}, \mathrm{K} \rangle$ is precisely the set of all "words" formed from the "alphabet" $\mathrm{H} \cup \mathrm{K}$.

Theorem 2.18 (Sylow's second and third theorems)**.** — *Let* G *be a finite group, of cardinality* $|\mathrm{G}| = p^{\alpha} m$*, with* $\alpha \in \mathbf{Z}_{>0}$ *and* $p \nmid m$*.*

(1) *If* H *is a* p*-subgroup of* G*, then there exists a Sylow* p*-subgroup* S*, with* $\mathrm{H} \subset \mathrm{S}$*.*
(2) *The Sylow* p*-subgroups of* G *are conjugates, and hence their number* k *divides* $n := p^{\alpha} m$*.*
(3) *We have* $k \equiv 1 \mod p$*, and hence* k *divides* m*.*

Proof. — If H is a p-subgroup and S is a Sylow p-subgroup, then there exists, thanks to Lem. 2.10, an element $a \in \mathrm{G}$ such that $\mathrm{H} \cap a\mathrm{S}a^{-1}$ is a Sylow p-subgroup of H. But since H itself is a p-group, we must have $\mathrm{H} = \mathrm{H} \cap a\mathrm{S}a^{-1}$. Consequently, $\mathrm{H} \subset a\mathrm{S}a^{-1}$, and of course $a\mathrm{S}a^{-1}$ is also a Sylow p-subgroup of G. Moreover, if H is also a Sylow p-subgroup, then $\mathrm{H} = a\mathrm{S}a^{-1}$. Thus we have shown that Sylow p-subgroups of G are conjugates. Consequently, if we consider the conjugation action of G on the set of all Sylow p-subgroups, then for a fixed Sylow p-subgroup, say S, the orbit $\mathcal{O}_{\mathrm{S}}$ would be the entire set of Sylow p-subgroups. Also by the orbit-stabilizer theorem $|\mathcal{O}_{\mathrm{S}}|$ divides the cardinality of G. We have thus proved the first two statements.

We now prove the third statement. Let X denote the set of Sylow p-subgroups of G. Fix a Sylow p-subgroup S, and let S act on X by conjugation : the map $\mathrm{S} \times \mathrm{X} \to \mathrm{X}$ is $(s, \mathrm{T}) \mapsto s \cdot \mathrm{T} := s\mathrm{T}s^{-1}$. By Lem. 2.12 we have $|\mathrm{X}| \equiv |\mathrm{X}^{\mathrm{S}}| \mod p$. So it remains to show that $|\mathrm{X}^{\mathrm{S}}| = 1$. First, observe that $\mathrm{S} \in \mathrm{X}^{\mathrm{S}}$ since for every $s \in \mathrm{S}$ obviously $s\mathrm{S}s^{-1} = \mathrm{S}$. We claim that S is the only element of X^{S}.

Suppose $\mathrm{T} \in \mathrm{X}^{\mathrm{S}}$. This means $s\mathrm{T}s^{-1} = \mathrm{T}$ for all $s \in \mathrm{S}$; in other words, T is normalized by S. We utilize an argument due to Frattini : consider the subgroup N of G generated by S and T. Then $\mathrm{S}, \mathrm{T} \subset \mathrm{N}$ are, *a fortiori*, Sylow p-subgroups of N. Now since N is generated by S and T, and S normalizes T, it implies, thanks to Ex. 2.17, that T is a normal subgroup of N. But since Sylow p-subgroups of N are conjugates, this forces $\mathrm{S} = \mathrm{T}$. □

Corollary 2.19. — *Let* S *be a Sylow* p*-subgroup of* G*. The following conditions are equivalent :*

- S *is normal in* G*.*
- S *is the unique Sylow* p*-subgroup of* G*.*
- $k = 1$*.*

One of the most common applications of the theorems of Sylow is *detection of simplicity* of groups.

Definition 2.20. — A non-trivial group G is called *simple* if the only normal subgroups of G are $\{e\}$ and G itself.

Example 2.21. — A group of order 63 is not simple.

Proof. — Since $63 = 3^2 \times 7$, we consider Sylow 7-subgroups of G. Let k be the number of all these Sylow 7-subgroups. Then $k \equiv 1 \mod 7$ and $k \mid 9$. This forces $k = 1$. Therefore, there is precisely one Sylow 7-subgroup, which is normal by Corl. 2.19. □

Example 2.22. — Let G be a group of order pqr where p, q, r are primes and $p > q > r$. Then G is not simple.

Proof. — For contradiction assume that G is simple. First, note that the number of Sylow p-subgroups, say k_p, satisfies : $k_p \equiv 1 \mod p$ and $k_p \mid qr$. Since $k_p \neq 1$ (otherwise the Sylow p-subgroup is normal by the above Corl. 2.19) hence k_p is either q or r or qr. But $k_p \equiv 1 \mod p$ implies $k_p > p$ and so $k_p = qr$.

Similarly, k_q the number of Sylow q-subgroups satisfies : $k_q \equiv 1 \mod q$ and $k_q \mid pr$. But then, assuming G to be non-simple, $k_q \neq 1$ implies $k_q > q$. Now since $k_p = qr$ and the Sylow p-subgroups intersect trivially, their union contains $qr(p-1)$ many non-identity elements. The remaining non-identity elements are $pqr-1-qr(p-1) = qr-1$ in number. Thus $k_q(q-1) \leq qr-1 = r(q-1)+(r-1) < r(q-1)+(q-1) = (q-1)(r+1)$ and hence $k_q < r+1$ i.e. $k_q \leq r < q$, a contradiction. Thus G must be simple. $\square$

3. Semi-direct products

We motivate the notion of semi-direct products with the example of *Euclidean motions*. Consider the Euclidean plane $\mathbf{R}^2$. For each vector $v \in \mathbf{R}^2$ we can define a map $\ell_v : \mathbf{R}^2 \to \mathbf{R}^2$ by $\ell_v(w) := v + w$. We consider the group $\mathrm{N} := \{\ell_v \mid v \in \mathbf{R}^2\}$ under the composition law, called the *group of translations*. The map $\ell_v \mapsto \ell_v(0) : \mathrm{H} \to \mathbf{R}^2$ identifies N as a subgroups of the additive group $(\mathbf{R}^2, +)$. Now consider the group

$$\mathrm{SO}(2) := \left\{ \begin{pmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{pmatrix} \mid \theta \in \mathbf{R} \right\}$$

of rotations about the origin in $\mathbf{R}^2$. Then SO(2) acts on $(\mathbf{R}^2, +)$ (and thus N) naturally by matrix multiplication, which corresponds to a homomorphism $\mathrm{SO}(2) \to \mathrm{GL}_2(\mathbf{R})$. Thus since each matrix in SO(2) is an invertible linear transformation we have an action of SO(2) on N by automorphism. Thus we have three things : the groups SO(2), N and a homomorphism $\mathrm{SO}(2) \to \mathrm{Aut}(\mathrm{N})$. We want to put a group structure on $\mathrm{N} \times \mathrm{SO}(2)$ which should give a concrete meaning to the phrase "group of translations *and* rotations" a.k.a. *Euclidean motions*. To be precise given two pairs (t, r) and (t', r') we want a group operation $*$ such that $(t, r) * (t', r')$ can also be written as (t'', r'').

Definition 3.1. — Let H and N be two groups and a homomorphism $\tau : \mathrm{H} \to \mathrm{Aut}(\mathrm{N})$, we define a group structure on the Cartesian product $\mathrm{N} \times \mathrm{H}$ as follows : for (n, h) and (n', h') we set

$$(n, h) \cdot (n', h') := (n(h \cdot n'), hh').$$

Then $\mathrm{N} \times \mathrm{H}$ equipped with this groups structure is called the *external semi-direct product of* N *by* H *relative to* τ. This group is denoted by $\mathrm{N} \rtimes_\tau \mathrm{H}$ or simply $\mathrm{N} \rtimes \mathrm{H}$ when the action is clear from the context.

Exercise 3.2. — Show that the above operation is indeed a group operation. In particular, show that the identity of the group is (1,1) and the inverse of an element (n, h) is $(h^{-1} \cdot n^{-1}, h^{-1})$.

Proposition 3.3. — *Let* H *and* N *be two subgroups of a group* G *satisfying :*

- N *is normal in* G*;*
- $\mathrm{H} \cap \mathrm{N} = \{1\}$*;*
- $\mathrm{G} = \mathrm{NH}$.

Then $\mathrm{G} \simeq \mathrm{N} \rtimes \mathrm{H}$*, and we say that* G *is* internal semi-direct product of N by H.

Proof. — Define an action $\tau : \mathrm{H} \to \mathrm{Aut}(\mathrm{N})$ by $\tau(h)(n) := hnh^{-1}$ for $h \in \mathrm{H}, n \in \mathrm{N}$. With respect to this we form the semi-direct product $\mathrm{N} \rtimes_\tau \mathrm{H}$. Define

$$\Phi : \mathrm{N} \rtimes_\tau \mathrm{H} \to \mathrm{G} \text{ by } (n, h) \mapsto nh.$$

Then it is clear that Φ is surjective. Also $nh = n'h'$ implies $n^{-1}n' = hh'^{-1} \in \mathrm{N} \cap \mathrm{H} = \{1\}$, and so $n = n', h = h'$. Thus Φ is injective as well. To check that it is a homomorphism we compute

$$\begin{aligned} \Phi((n, h) \cdot (n', h')) &= \Phi(n(hn'h^{-1}), hh') \\ &= n(hn'h^{-1})hh' \\ &= (nh)(n'h') \\ &= \Phi(n, h)\Phi(n', h'). \end{aligned}$$

This completes the proof. $\square$

Remark 3.4. — The above proposition essentially says that internal semi-direct products can be viewed as external semi-direct products. The reverse point of view is also useful.

Exercise 3.5. — Let N, H be two groups and $\tau : \mathrm{H} \to \mathrm{Aut}(\mathrm{N})$ a group homomorphism. Form the semi-direct product $\mathrm{N} \rtimes_\tau \mathrm{H}$. Consider the maps

$$\mathrm{N} \hookrightarrow \mathrm{N} \rtimes_\tau \mathrm{H}\,, \quad \mathrm{H} \hookrightarrow \mathrm{N} \rtimes_\tau \mathrm{H}$$

given by $n \mapsto (n,1)$ and $h \mapsto (1,h)$ respectively. Show that these two maps are group homomorphisms. Denote their images by $\bar{\mathrm{N}}$ and $\bar{\mathrm{H}}$ respectively. Show that $\mathrm{N} \rtimes_\tau \mathrm{H}$ is the internal semi-direct product of $\bar{\mathrm{N}}$ by $\bar{\mathrm{H}}$ i.e. show that

- $\bar{\mathrm{N}}$ is normal in $\mathrm{N} \rtimes_\tau \mathrm{H}$;
- $\bar{\mathrm{N}} \cap \bar{\mathrm{H}} = \{(1,1)\}$;
- $\mathrm{N} \rtimes_\tau \mathrm{H} = \bar{\mathrm{N}}\bar{\mathrm{H}}$.

Exercise 3.6. — Let $\mathrm{N} \rtimes_\tau \mathrm{H}$ be a semi-direct product of N by H as in the previous exercise. Show that the following are equivalent :

(1) τ is the trivial homomorphism i.e. $\tau(h) = \mathrm{id}_\mathrm{N}$ for every $h \in \mathrm{H}$;
(2) the subgroup $\bar{\mathrm{H}}$ is normal in $\mathrm{N} \rtimes_\tau \mathrm{H}$;
(3) the group operation on $\mathrm{N} \rtimes_\tau \mathrm{H}$ is the same as that of direct product.

Hint : Show that $1 \iff 2$ and $1 \iff 3$.

We now give a different characterization of semi-direct product in terms of *exact sequences*. A sequence of groups and homomorphisms

$$\cdots \to \mathrm{G}_{i-1} \xrightarrow{\phi_{i-1}} \mathrm{G}_i \xrightarrow{\phi_i} \mathrm{G}_{i+1} \to \cdots$$

is called *exact* if for every i we have $\mathrm{Ker}\,\phi_i = \mathrm{Im}\,\phi_{i-1}$. Thus in particular the sequence

$$1 \to \mathrm{N} \xrightarrow{i} \mathrm{G} \xrightarrow{p} \mathrm{H} \to 1$$

is exact if : i is injective, p is surjective, and $\mathrm{Ker}\,p = \mathrm{Im}\,i$. The above sequence is also often called a *short exact sequence.*

Proposition 3.7. — *Consider the short exact sequence of groups*

$$1 \to \mathrm{N} \xrightarrow{i} \mathrm{G} \xrightarrow{p} \mathrm{H} \to 1.$$

The following are equivalent :

(1) *There exists a "lift"* $\bar{\mathrm{H}}$ *of* H *i.e. a subgroup* $\bar{\mathrm{H}} \subset \mathrm{G}$ *such that* $p|_{\bar{\mathrm{H}}} : \bar{\mathrm{H}} \xrightarrow{\sim} \mathrm{H}$ *is an isomorphism of groups.*
(2) p *has a "section" i.e.there is a homomorphism* $s : \mathrm{H} \to \mathrm{G}$ *such that* $p \circ s = \mathrm{id}_\mathrm{H}$*. We also often say that the sequence is split if* p *has a section.*

If one of the above two conditions hold , then $\mathrm{G} \simeq \mathrm{N} \rtimes \mathrm{H}$.

Proof. — ($1 \implies 2$) This is clear : define $s := \iota_{\bar{\mathrm{H}}} \circ (p|_{\bar{\mathrm{H}}})^{-1}$ where $\iota_{\bar{\mathrm{H}}} : \bar{\mathrm{H}} \hookrightarrow \mathrm{G}$ is the inclusion homomorphism.

($2 \implies 1$) If $s : \mathrm{H} \to \mathrm{G}$ is a homomorphism such that $p \circ s = \mathrm{Id}_\mathrm{H}$ then set $\bar{\mathrm{H}} := s(\mathrm{H})$. Consider the map $p|_{\bar{\mathrm{H}}} : \bar{\mathrm{H}} \to \mathrm{H}$. We claim that this is bijective. First, note that for any $h \in \mathrm{H}$ we have $h = p(s(h)) \in p(\bar{\mathrm{H}})$ and so we have surjectivity. On the other hand, suppose $x = s(h) \in \bar{\mathrm{H}}$ is such that $p(x) = 1$; but $p(x) = p(s(h)) = h$, and so $h = 1$, consequently, $x = s(h) = s(1) = 1$.

Finally, we show that in the above situations we have $\mathrm{G} \simeq \mathrm{N} \rtimes \mathrm{H}$. First note that since s is injective $\bar{\mathrm{H}} \simeq \mathrm{H}$. Set $\bar{\mathrm{N}} := i(\mathrm{N}) \simeq \mathrm{N}$: since i is injective. We show that G is internal semi-direct product of $\bar{\mathrm{N}}$ by $\bar{\mathrm{H}}$. We easily observe that $\bar{\mathrm{N}}$ is normal since $\bar{\mathrm{N}} = i(\mathrm{N}) = \mathrm{Ker}\,p$. Now if $x \in \bar{\mathrm{N}} \cap \bar{\mathrm{H}}$ then write $x = s(h)$ for some $h \in \mathrm{H}$; but $p(x) = 1$ i.e. $p(s(h)) = h = 1$ and so $x = 1$. Thus $\bar{\mathrm{N}} \cap \bar{\mathrm{H}} = \{1\}$. Finally, take any $g \in \mathrm{G}$ and consider $h := p(g) \in \mathrm{H}$. Then $gs(h^{-1}) \in \mathrm{Ker}\,p = \bar{\mathrm{N}}$ and hence $g \in \bar{\mathrm{N}}s(h) \in \bar{\mathrm{N}}\bar{\mathrm{H}}$. Thus we have shown $\mathrm{G} = \bar{\mathrm{N}}\bar{\mathrm{H}}$. □

Exercise 3.8. — Given an external semi-direct product $\mathrm{N} \rtimes \mathrm{H}$ show that the canonical short exact sequence

$$1 \to \mathrm{N} \to \mathrm{N} \rtimes \mathrm{H} \to \mathrm{H} \to 1$$

is split.

Semi-direct products can also be used to describe groups of order pq where p and q are distinct primes.

Theorem 3.9. — *Let p and q be prime and $p < q$.*

(1) *If $p \nmid (q-1)$, then every group of order pq is cyclic.*
(2) *If $p \mid (q-1)$, there there exists up to isomorphism two groups of order pq, namely, the cyclic groups $\mathbf{Z}/pq$ and the non-commutative semi-direct product $\mathbf{Z}/q \rtimes \mathbf{Z}/p$.*